Syracuse University
School of Information Studies

Week 4
Samples and Populations

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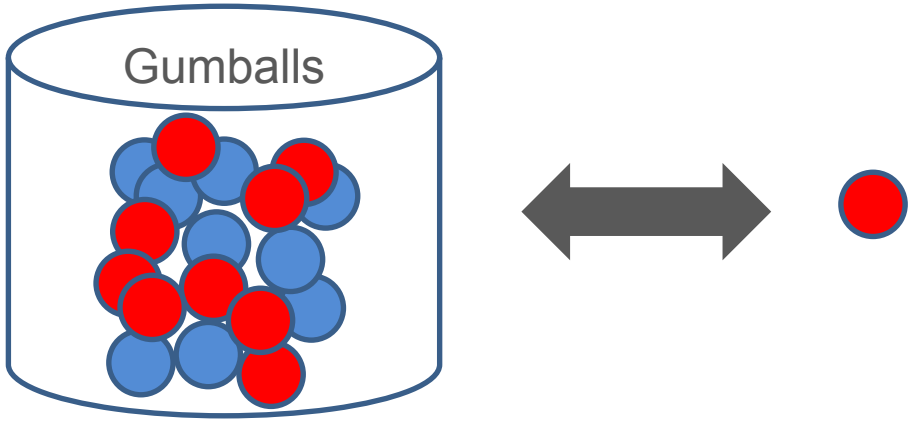
Population and Sample

- A **population** in research means the entire group of people or things that you're interested in studying.
- **Example:** If you're studying iPhone users, the population could be all iPhones currently being used around the world.
- Even though we define a population clearly, we usually can't study every single member of it, because it's too big, expensive, or unavailable.
- Instead of studying the whole population, researchers take a **sample (**smaller group that represents the population) and try to make conclusions about the population from it.
- For **example:** Instead of testing every iPhone, you might survey 1,000 users from different countries.

Sampling Process



- **Random sampling** is when every element has an equal chance of being drawn. best represents the population it was drawn from.
- **Replacement** is when we put the element back after we have drawn it but before we draw the next random element.

The Law of Large Numbers:

- If you repeat a random experiment many, many times, the average result will get closer and closer to the true average.

- **Example**:
  - Imagine flipping a fair coin (50% chance of heads, 50% tails): If you flip it just 2 times, you might get 2 heads. That's 100% heads, which is not the real probability.
  - If you flip it 10 times, you might get 6 heads, that's 60%.
  - But if you flip it 1,000 times, you'll start getting closer to 50% heads, the true rate.

Central Limit Theorem:

- When you take many samples from a population and calculate their means, those means will form a bell-shaped curve (normal distribution).
- The mean of the sample means, gets close to the population mean

Percentiles and Quartiles

- These terms describe positions within a data set.
- **Quartile** divides data into 4 equal parts.
- **Percentile** divides data into 100 equal parts.
- The **median** is both the 50th percentile and the 2nd quartile.

Coding

```
numbers <- c(10, 20, 30)
mean(sample(numbers, size = 50, replace = TRUE))
```

- Randomly selects 50 numbers.
- Each number comes from {10, 20, 30}
- **replace = TRUE** means values can repeat
- **mean(...):** Calculates the average of those 50 sampled numbers

Output: Produces one numeric value. The value will **be close to 20**. It changes slightly each time due to randomness

```
data <- c(10, 20, 30)
mean(replicate(1000, mean(sample(data, size = 3, replace = FALSE))))
```

- Randomly selects 3 numbers.
- Each number comes from {10, 20, 30}
- **replace = FALSE** means values can't repeat
- **mean(...):** Calculates the average of those 3 sampled numbers
- **replicate(1000, ...):** Repeats this 1000 times

Output: Produces one numeric value. The value will be **always 20**.

```
numbers <- c(10, 20, 30)
sample(numbers, size = 1, replace = FALSE)
```

- **sample()** randomly selects values
- **size = 1** means pick exactly one number
- **replace = FALSE** means no repetition (irrelevant here since only one value is picked)

Output: The output is one randomly chosen number. **Possible outputs: 10, 20, or 30.** **Result changes each time you run it.**

rbinom()

- R has the capability to generate random numbers in a variety of configurations.
- When working with events that have two states, such as heads and tails, the best distribution to use is the **binomial distribution**.
- In R, the "**rbinom()**" command generates random trials of the binomial distribution.

```
rbinom(n=8, size=1, prob=0.5)

## [1] 1 0 0 0 0 0 0 0
```

n = 8: Generate 8 values.
size = 1: Each value is the result of 1 trial (like flipping 1 coin).
prob = 0.5: The chance of getting a 1 (success) is 50%.
Each value will be : 1 = success (e.g., heads), 0 = failure (e.g., tails).

```r
set.seed(10)
rbinom(n=8,size=1,prob=0.5)
```

```
## [1] 1 0 0 1 0 0 0 0
```

```r
factor(rbinom(n=8,size=1,prob=0.5),labels=c("red","blue"))
```

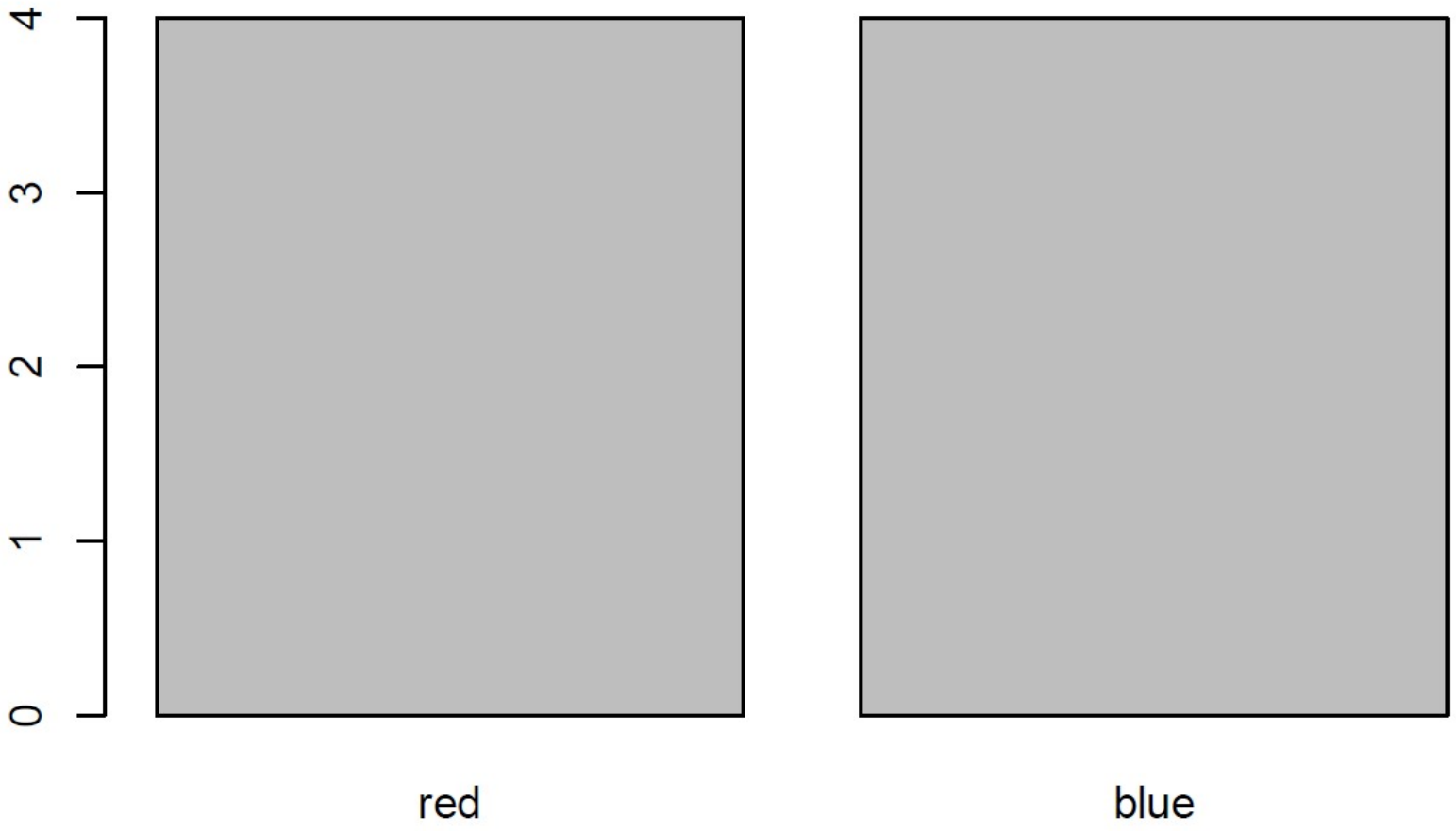
```
## [1] blue red  blue blue red  blue red  red
## Levels: red blue
```

- **set.seed(10):** Sets the seed for R's random number generator, so if someone else runs the same code, they will get the exact same output.
- **factor(rbinom(n=8, size=1, prob=0.5), labels = c("red", "blue")):** This line does another set of 8 coin flips then replaces the numbers (0 and 1) with labels: 0 . "red", 1 . "blue"
- So if the random values were: **0 0 1 1 0 1 0 0**, You would get: **red red blue blue red blue red red**
- **Levels: red blue:** It turns numeric outcomes into category names (called a factor in R).

Tabulate and Plot:

```
barplot(table(factor(
    rbinom(n=8,size=1,prob=0.5),
            labels=c("red","blue"))))
```

The **table()** function counts how many times each category appears.
So, the output means: 4 outcomes were red; 4 outcomes were 1



Finding the Histogram

```
hist(replicate(2000, mean(sample(stateDF$population,
                size=16,replace=TRUE)),simplify=TRUE),
       breaks=30,
       main ="Distribution of sampled means",
       xlab="Sampled population means")
```

sample(stateDF$population, size = 16, replace = TRUE)

- Randomly picks **16 values**
- From the population data
- Values **can repeat**

mean(sample(...))

Computes the **average** of those 16 values

replicate(2000, ...)

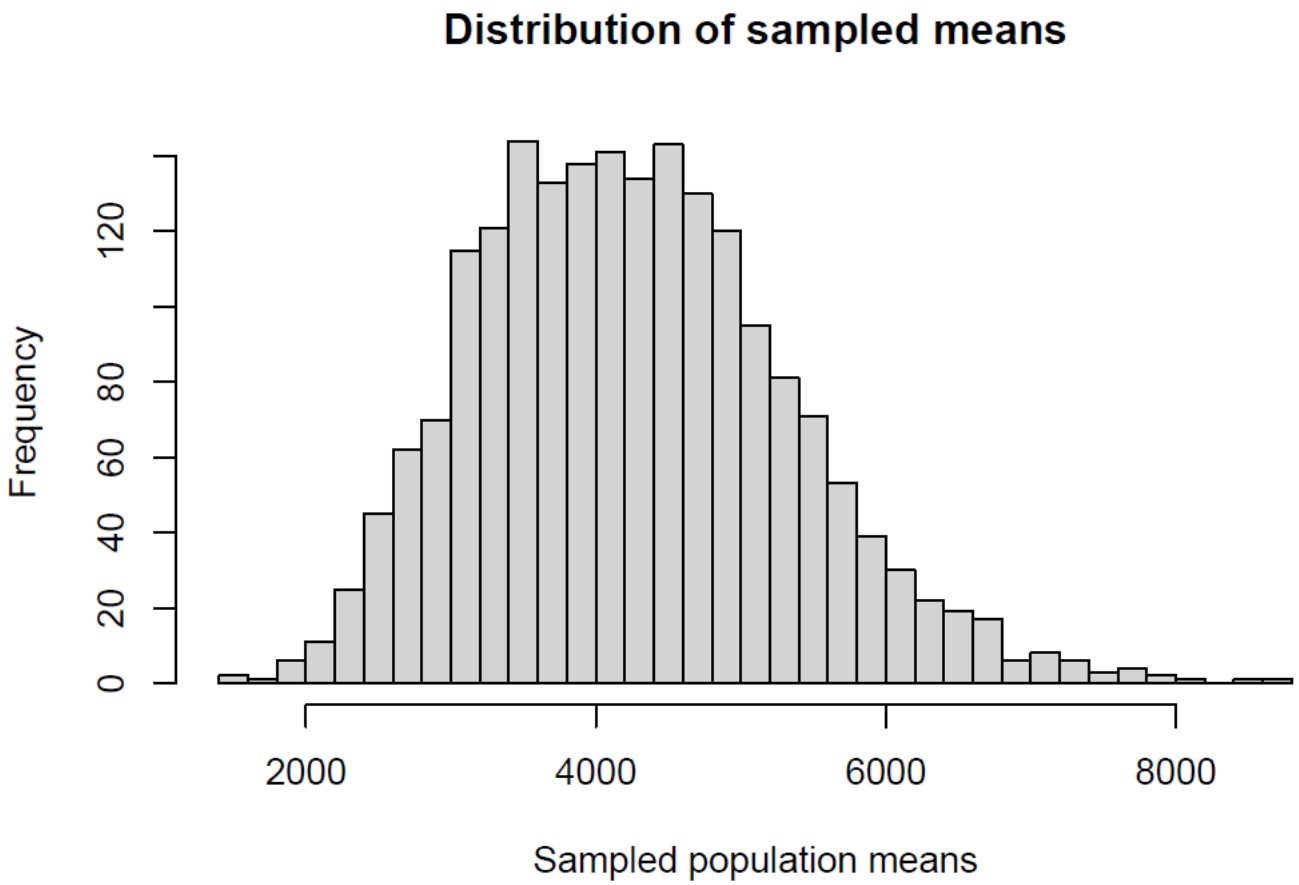
- Repeats the sampling-and-mean process **2000 times**
- Produces **2000 means**

simplify = TRUE

- Stores the 2000 means as a **numeric vector**

hist(...)

Draws a **histogram** of the 2000 means



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The End